ALFARIZI

Software Engineer | Backend | Full-Stack | Python & AI

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PROFESSIONAL SUMMARY

Entry-level Software Engineer with hands-on experience building backend systems, full-stack applications, distributed services, and AI-powered applications through professional, academic, and independent projects. Experienced with Java, Spring Boot, Quarkus, React, Next.js, Python, FastAPI, PostgreSQL, Redis, Docker, and CI/CD. Strong interest in backend engineering, full-stack development, AI applications, distributed systems, and API-driven software.

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, C

Backend Development: Spring Boot, Quarkus, Spring Cloud Gateway, FastAPI, Express.js, REST APIs

Frontend Development: React, Next.js, Vite, Tailwind CSS, Web Speech API

Database & Data: PostgreSQL, Redis, ChromaDB

AI & Application Development: RAG, vector search, document processing, Gemini, PyMuPDF, AI-powered applications

DevOps & Infrastructure: Docker, Docker Compose, Git, GitHub Actions, Linux

Software Engineering Tools: OpenAPI/Swagger, ArchUnit, Testcontainers, TanStack Query

PROFESSIONAL EXPERIENCE

Lead Intern — IT & Business Operations
PT Resika Cemerlang (Pasim Group), Bandung | 2025 – Sep 2026

- Led a 5-person intern team supporting daily IT operations, hardware diagnostics, and workstation SLA compliance.
- Managed OS security patching, software configuration, and antivirus definition updates across corporate laptops.
- Validated technical specifications across thousands of medical device asset records, maintaining database integrity and audit readiness.

LEADERSHIP & ORGANIZATIONS

Academic Staff — HIMAMI

- Coordinated academic development activities, peer-to-peer coding support, and technical learning initiatives.

SELECTED PROJECTS

ReLoop — Enterprise Circular-Commerce Platform & Escrow Architecture
Java 17, Quarkus 3.15 LTS, Next.js 14, PostgreSQL 16, Redis 7 | 2026

- Architected a modular monolith across 25 bounded contexts, enforcing module boundaries with ArchUnit and asynchronous processing through a transactional outbox worker.
- Engineered a double-entry escrow ledger and transaction-safe checkout leases using PostgreSQL pessimistic locking (SELECT ... FOR UPDATE) to prevent concurrent inventory conflicts.
- Integrated Next.js with TanStack Query and Redis catalog caching for responsive product browsing and inventory-aware checkout.
- Demo: reloop.biz.id GitHub: github.com/a1fariz/ReLoop

ApexGrid — F1 Race Ticket Reservation & Microservices Platform
Java 17, Spring Boot 3.2, Spring Cloud Gateway, PostgreSQL, React 18, Docker | 2026

- Architected a 3-service distributed system consisting of Gateway, User, and Race services.
- Implemented stateless JWT-based RBAC security using Spring Cloud Gateway.
- Synchronized ticket inventory with PostgreSQL PL/pgSQL triggers and an automated 30-minute reservation-expiry worker.
- Built a React-based admin analytics panel with OpenAPI/Swagger documentation and containerized the platform using Docker Compose.
- GitHub: github.com/a1fariz/ApexGrid

StudyMate AI — Asynchronous RAG Document Knowledge Assistant
Python 3.12, FastAPI, ChromaDB, Gemini 2.0 Flash, Docker | 2026

- Built an asynchronous RAG pipeline that parses textbook PDFs using PyMuPDF, performs vector search with ChromaDB, and returns exact page citations.
- Integrated Gemini 2.0 Flash for summaries, quizzes, and 3D flashcards.
- Implemented a 60 requests/minute token-bucket rate limiter.
- GitHub: github.com/a1fariz/StudyMateAI

Vanguard IELTS — Academic Preparation & Spaced-Repetition Platform
React 19, TypeScript, Vite, Tailwind CSS v4, Web Speech API | 2026

- Developed an IELTS preparation suite covering Listening, Reading, Writing, and Speaking across CEFR A1–C1 levels.
- Implemented a SuperMemo-inspired spaced-repetition vocabulary engine.

- Built Shadowing Studio functionality using native Web Speech synthesis.
- Demo: vanguard-ielts.vercel.app GitHub: github.com/a1fariz/Vanguard-IELTS

EDUCATION

Universitas Nasional PASIM
Associate Degree in Informatics Management (D3) | GPA: 3.78/4.00

Academic Status: Coursework completed; currently preparing for final thesis defense (sidang).

PUB Intensive Technical Scholarship
Logic & C, Data Structures, Relational Databases, Advanced Java Development, and React Development | 2024–Present

CERTIFICATIONS & TECHNICAL TRAINING

- Python for Everybody Specialization — University of Michigan, 2026
- Programming for Data Science with Python — HarvardX, 2025
- Certified Entry-Level Python Programmer (PCEP) — Python Institute, 2025
- Introduction to Cloud Computing — IBM, 2026
- Intensive React Front-End Development Training — PUB Programming Training Center, PASIM National University, Sep 2026
- Fundamental Back-End Programming Training — Java Programming Language — PUB Programming Training Center, PASIM National University, Jan 2026
- Git and GitHub Training — PUB Programming Training Center, PASIM National University, Jan 2026
- Web Programming Training — HTML, CSS, JavaScript — PUB Programming Training Center, PASIM National University, Aug 2025
- Database Training — PUB Programming Training Center, PASIM National University, May 2025
- Data Structure Training — PUB Programming Training Center, PASIM National University, May 2025

ADDITIONAL STRENGTHS

- Backend API development
- Full-stack web application development
- Microservices and distributed systems
- Database-driven application development
- AI and RAG application development
- Authentication and authorization
- REST API integration
- Dockerized application environments
- Technical problem solving and debugging
- Git-based collaborative development